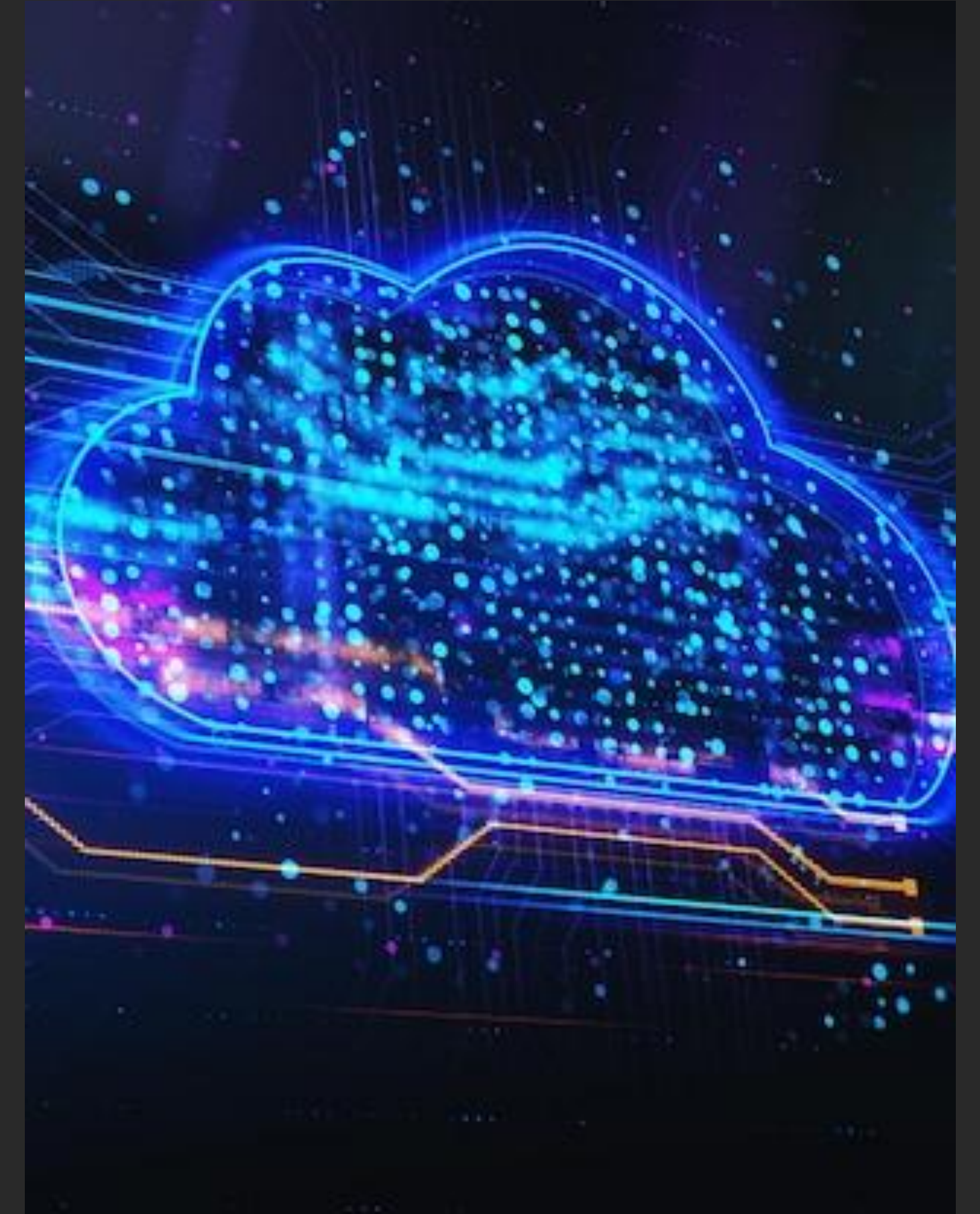


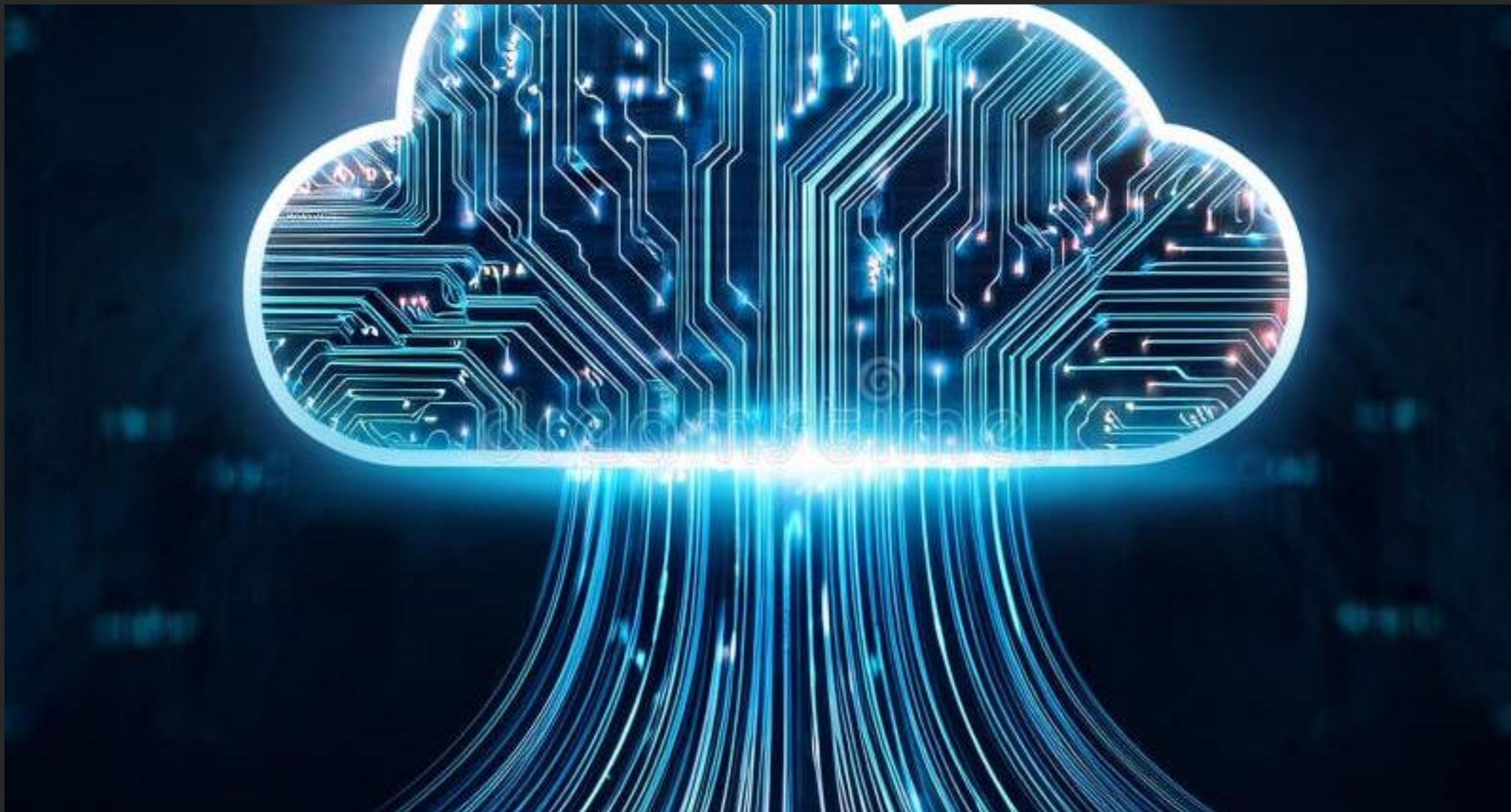
Secure Student Management System

FutureTech Academy | Cloud-Based Solution

Group Presentation | APDA 301



Presentation Overview



- 01 Project Overview
- 02 System Features
- 03 Live Demo
- 04 Deployment & Testing

Secure Student Management System on Azure

01

Project Overview

Understanding the problem, solution, and technology behind our secure student management system



Problem & Solution

THE PROBLEM

- Manual student record management leads to **data loss** and inefficiency across departments
- **Limited access control** makes it difficult to manage who can view or edit student records
- No centralized system results in **fragmented data** scattered across different platforms

OUR SOLUTION

- **Secure cloud-based system** built on Azure with enterprise-grade security
- **Centralized student data** with easy access anytime, anywhere via web browser
- **Role-based access control** ensures data security with admin and viewer permissions

Technology Stack

Built with modern Microsoft Azure cloud technologies



.NET Core

Web Layer handling UI and business logic with modern MVC architecture



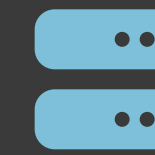
Azure Cosmos DB

Database Layer storing student records with global distribution and low latency



Azure Blob Storage

Storage Layer for student profile images in secure 'student-images' container



Azure App Services

Hosting platform with auto-scaling, deployment slots, and monitoring

Integration: `profileImageUrl` links database records to blob storage images

02

System Features

Authentication, security, user roles, and core system capabilities



Authentication & Security

Enterprise-grade security for student data protection



OAuth 2.0 Login

Secure authentication via Google and GitHub with industry-standard OAuth 2.0 protocol



Role-Based Access Control

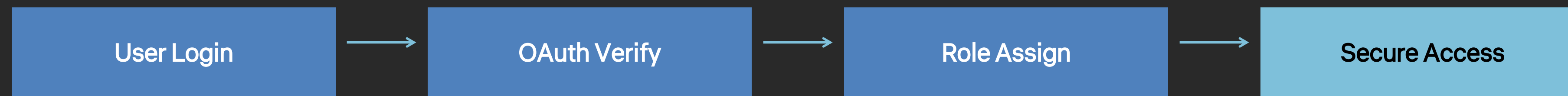
RBAC system ensures users only access features matching their assigned permissions



SAS Token Security

Shared Access Signature tokens provide secure, time-limited access to image files

Authentication Flow



System Features

Complete CRUD operations with file management

- **CRUD Operations**

Create, read, update, and delete student records with intuitive interface

- **File Validation**

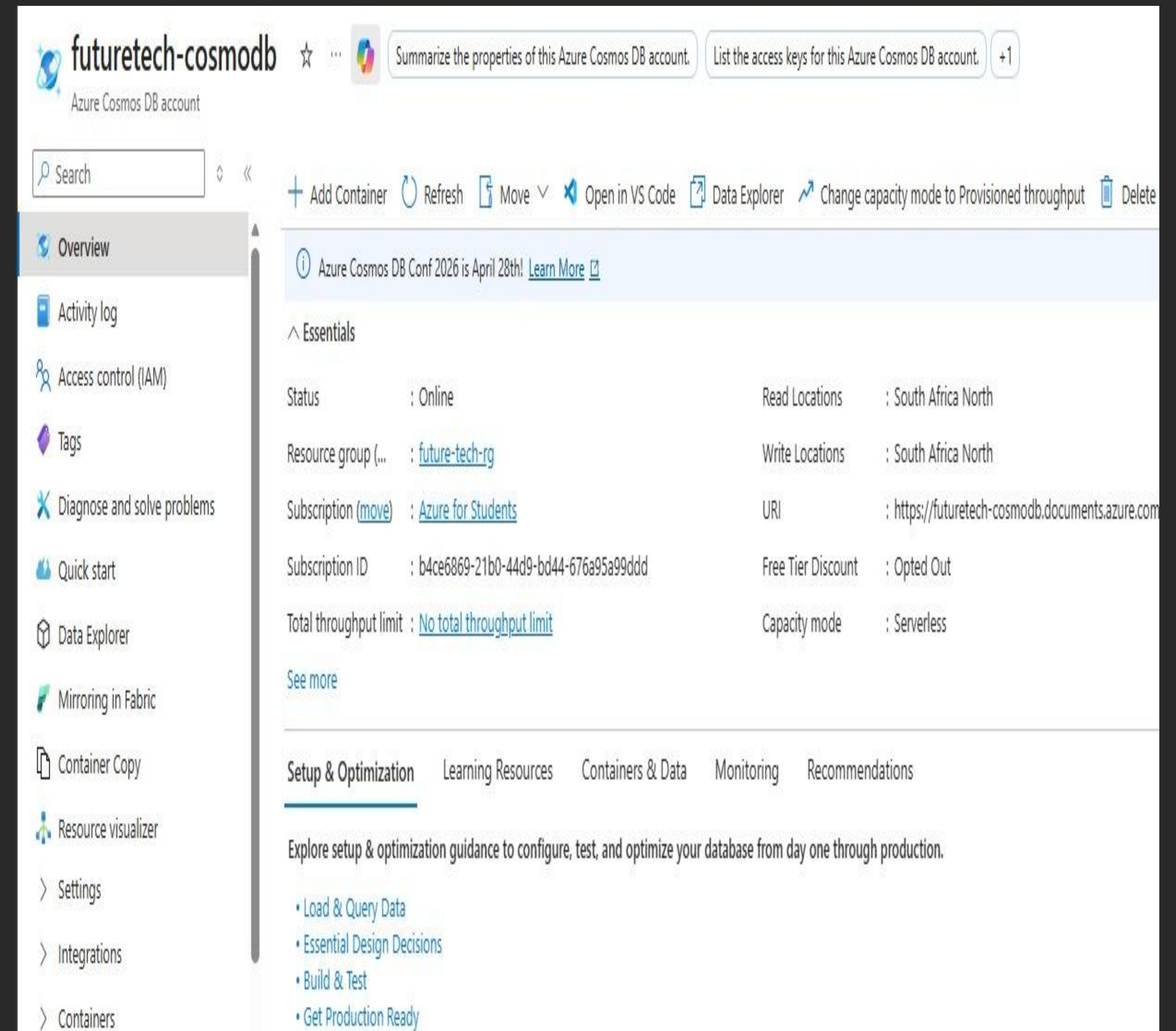
Only JPEG and PNG image formats accepted for student profile photos

- **User-Friendly Dashboard**

Clean, responsive interface with sidebar navigation and search functionality

- **Azure App Services**

Deployment slots enable seamless updates with zero downtime



User Roles & Access

Two-tier role system for secure data management



ADMINISTRATOR

- ✓ Full CRUD access to all student records
- ✓ Add new students with profile images
- ✓ Edit and delete student records
- ✓ Activate/deactivate student status
- ✓ Search by ID, Name, or Email
- ✓ Access system settings and logs



VIEWER

- ✓ View all student records (read-only)
- ✓ Search students by ID, Name, Email
- ✓ View read-only access banner
- ✓ Navigate dashboard and sidebar

New users are granted Viewer access by default. Admin role must be assigned.

03

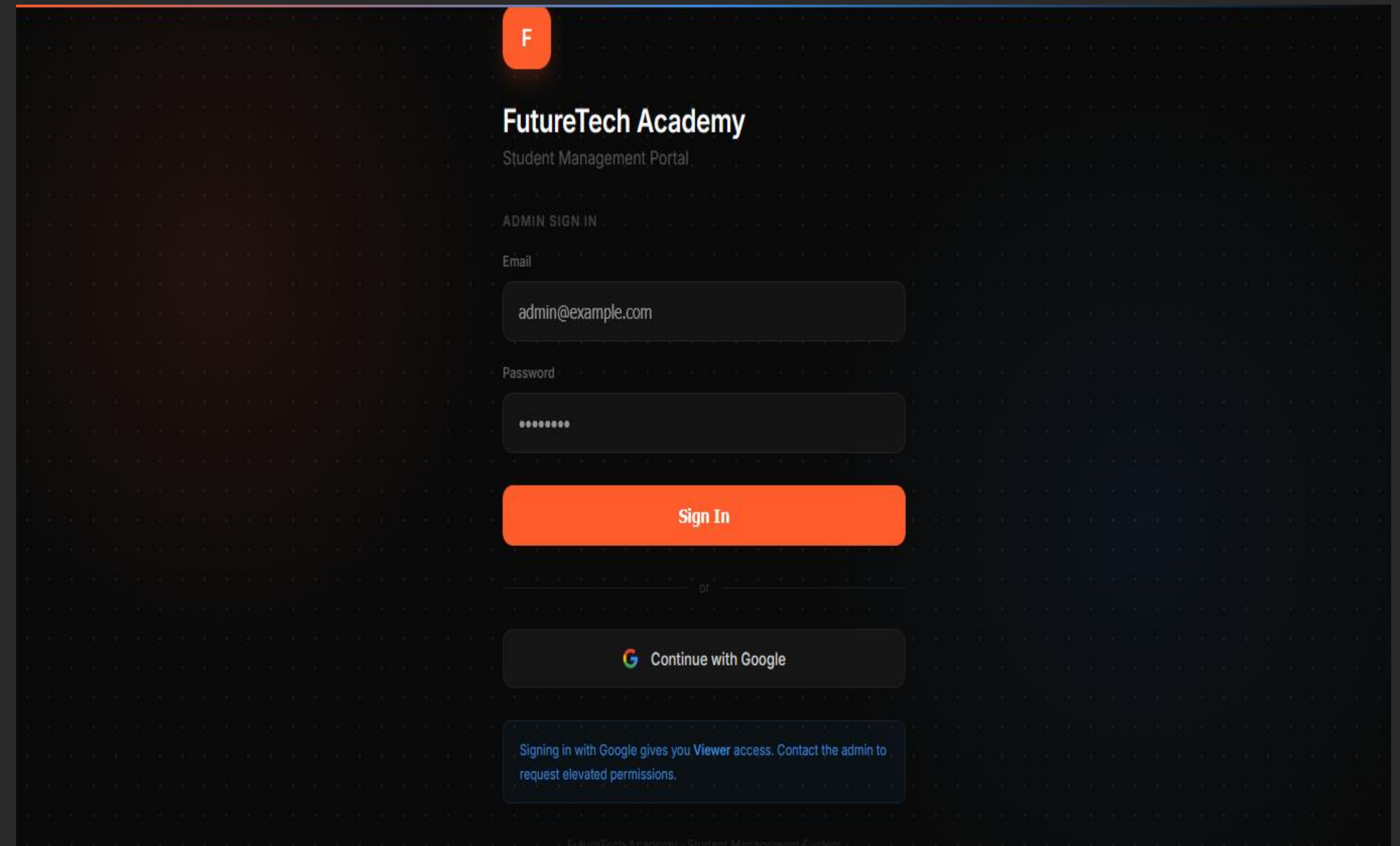
Live Demo

Visual walkthrough of the system - login, dashboard, student management, and backend

Login Screen

The system supports multiple authentication methods:

- First-time admin account setup
- Email and password sign-in
- Google OAuth integration

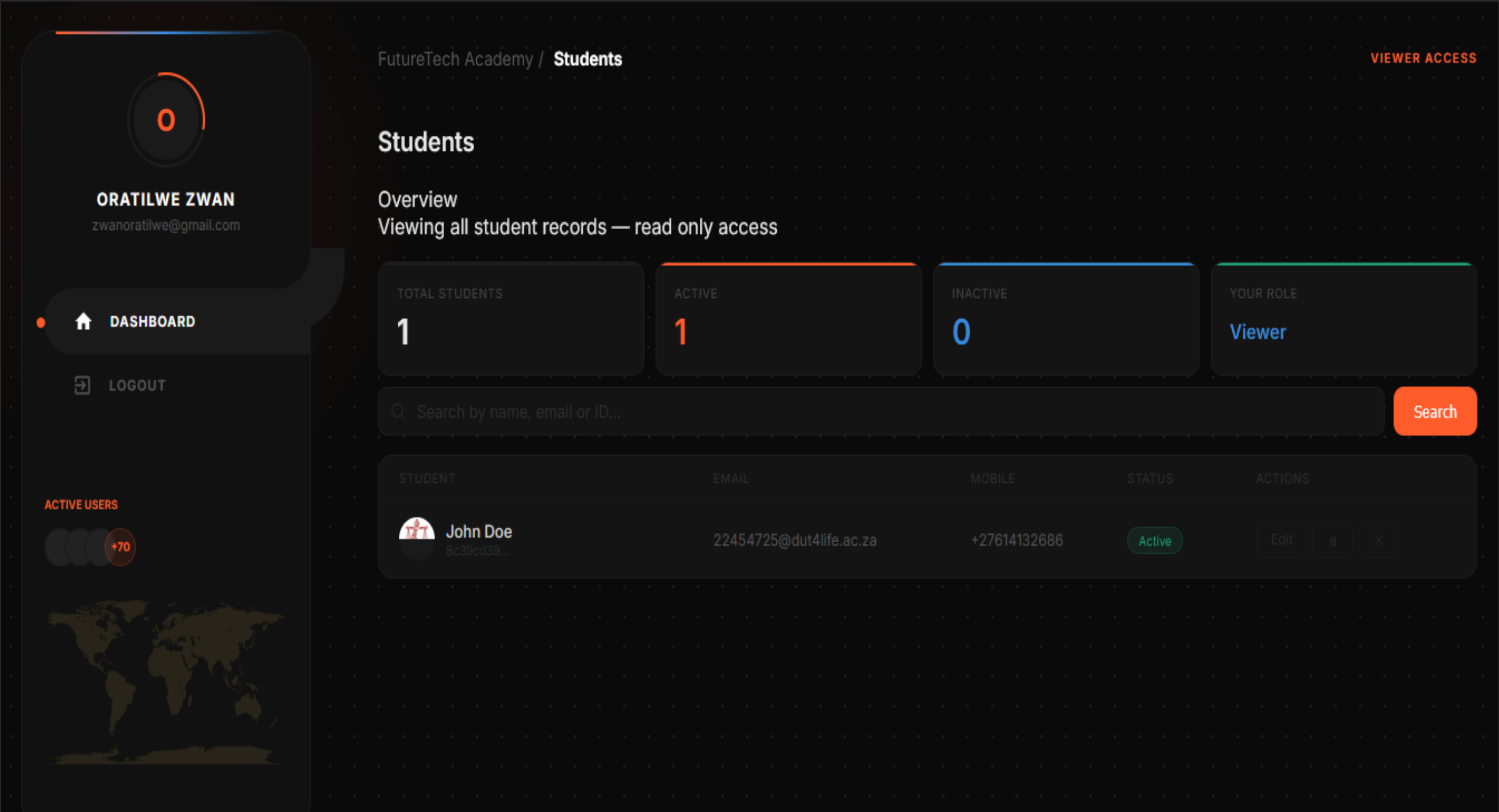


Tip: New users are granted **Viewer Access** by default. If you are an Admin, you will be granted full access after authentication.

Dashboard View

The main dashboard provides an overview of all student records:

- Student list with search and filter
- Sidebar navigation panel
- Read-only access banner for viewers



Tip: The sidebar contains the **User Profile**, navigation between Dashboard and Students sections, and logout functionality.

Add Student Form

The add student form captures all required information:

- Personal Info (First Name, Last Name)
- Contact details (Email, Mobile)
- Enrollment status (Active/Inactive)
- Profile image upload (JPEG/PNG)

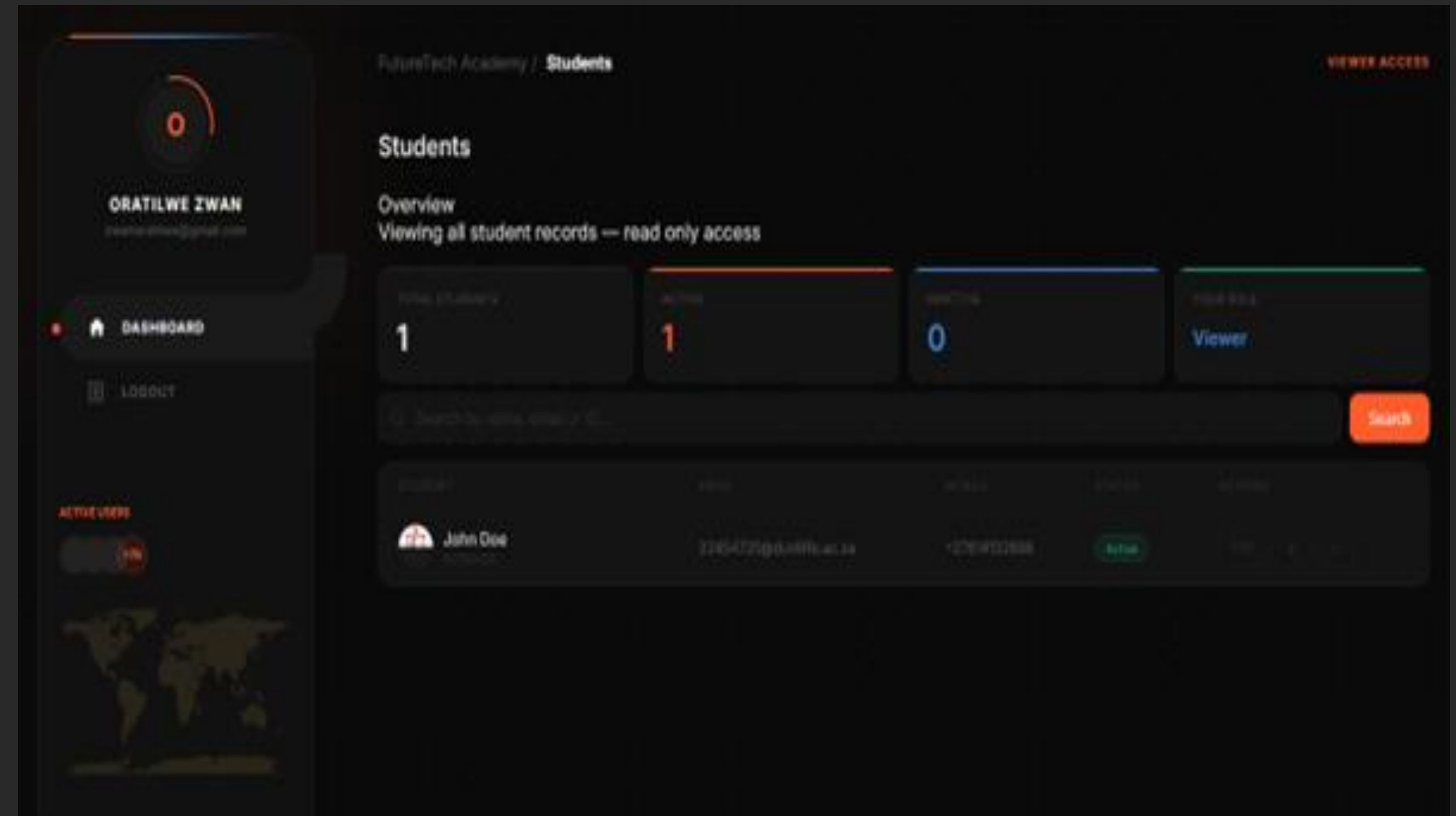
The screenshot shows a web application interface for adding a new student. On the left is a sidebar with a user profile for 'JANE DOE' (email: futuretech.admin.nextgen@gmail.com) and navigation links for DASHBOARD, STUDENTS (active), and LOGOUT. Below the sidebar, it shows 'ACTIVE USERS' as 70 and a world map. The main header displays 'FutureTech Academy / Add Student' and the user role 'ADMINISTRATOR'. The main content area has a title 'Add New Student' with a subtext 'Fill in the student details and upload a profile picture' and a 'Back to Students' link. A red error message states 'The ProfileImage field is required.' The form is divided into two columns. The left column, titled 'PERSONAL INFO', contains fields for First Name (Sipho), Last Name (Khumalo), Email Address (22145678@du4life.ac.za), and Mobile Number (+27 71 234 5678). The right column, titled 'ENROLMENT', has an 'Enrolment Status' dropdown set to 'Active'. Below this is a 'PROFILE IMAGE' section with a dashed box and text 'Click to upload photo. JPEG or PNG — max 5MB'. At the bottom, a note says 'Fields marked * are required', and there are 'Cancel' and 'Save Student' buttons.

Tip: Click **Save Student** to commit the record. A confirmation message appears when the student is added successfully.

Admin Actions

Administrators can perform full record management:

- Edit student records inline
- Pause/Activate student status
- Delete with confirmation dialog
- Search by ID, Name, or Email

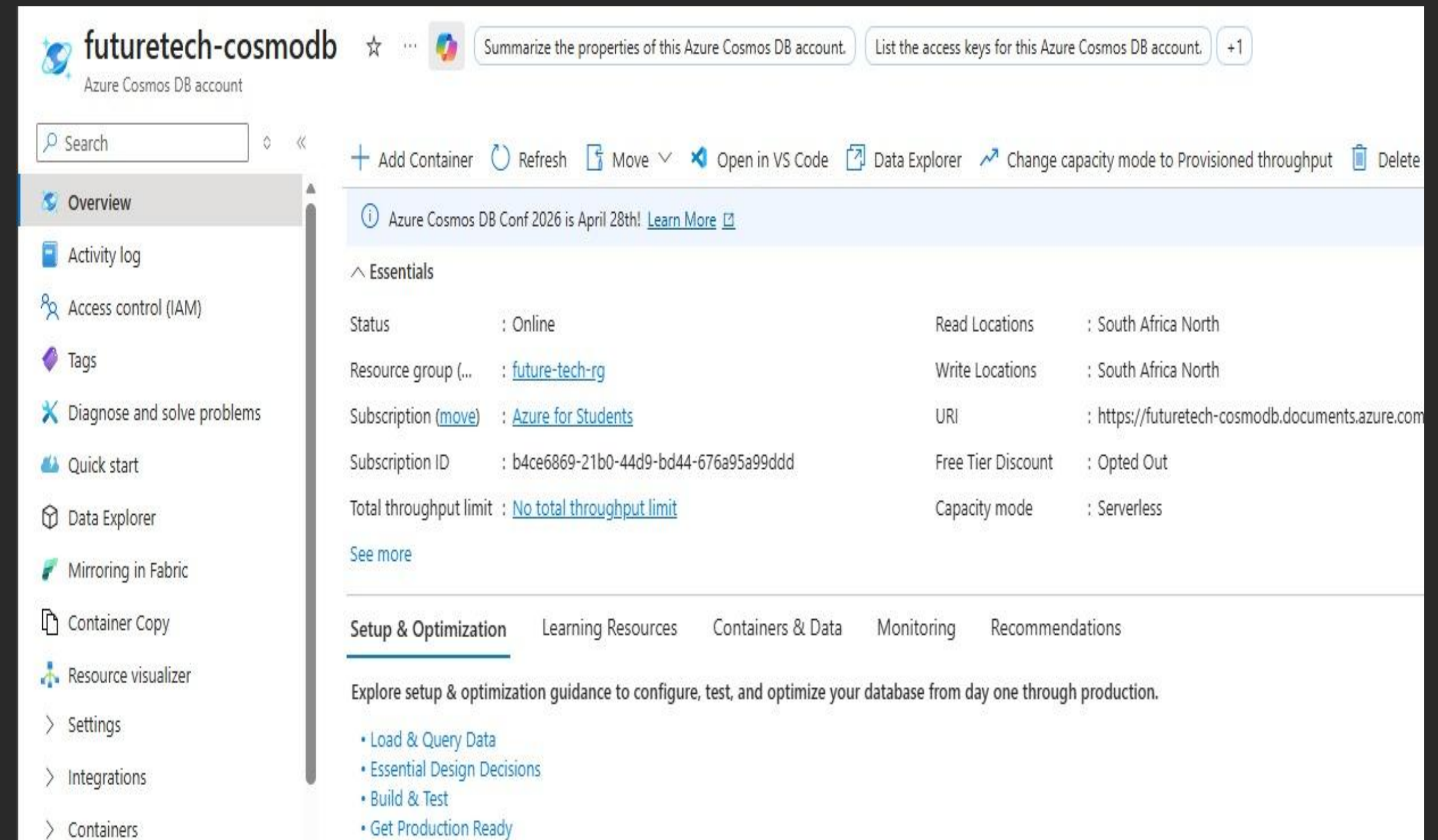


Warning: Deleting a student is **permanent** with no rollback. A confirmation dialog appears before deletion.

Azure Backend

The Azure cloud infrastructure powering the system:

- Cosmos DB student records collection
- Blob Storage 'student-images' container
- Azure Portal resource overview
- Application Insights monitoring

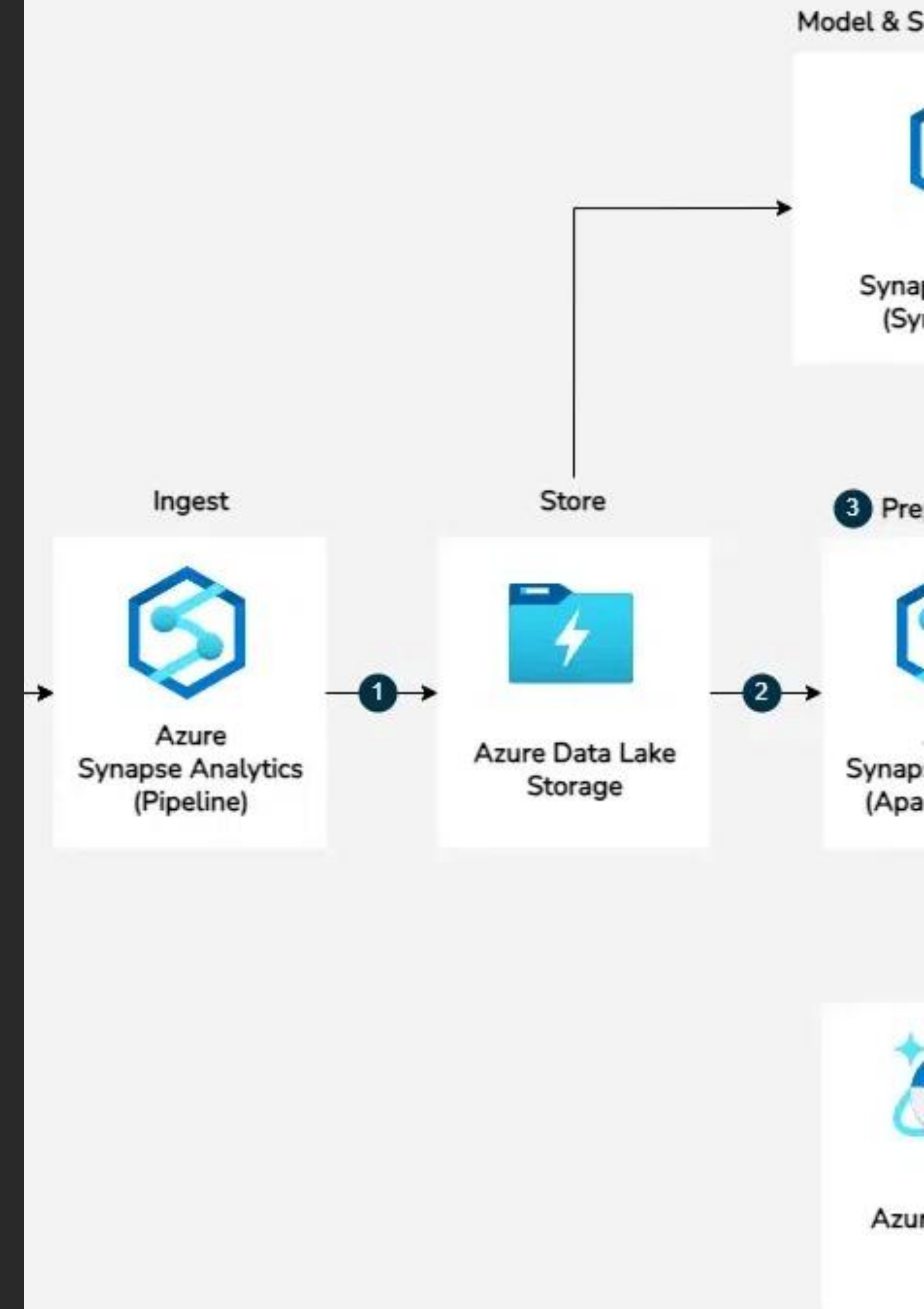


Integration: The `profileImageUrl` field in Cosmos DB links each student record to their image in Blob Storage.

04

Deployment & Testing

Live deployment, testing strategy, and project value



Deployment & Monitoring



Azure App Services

Hosted on Azure with deployment slots for seamless updates and zero downtime



Auto-Scaling Enabled

Automatically scales resources based on traffic demand to ensure performance



Application Insights

Real-time monitoring and diagnostics via Azure Application Insights

Project Links

[Live Project on Azure](#)

[GitHub Source Code](#)

Testing Strategy

Comprehensive testing across three key dimensions



Unit Testing

Tests for all CRUD operations and authentication flows to ensure core functionality works correctly



Security Testing

Validates OAuth authentication and RBAC permissions to ensure unauthorized access is blocked



Performance Testing

Measures response times and system behavior under load to ensure smooth user experience

Unit Test Results

Test Explorer			
Ready			
Test	Duration	Traits	Error Message
FutureTech.Test (15)	1,3 sec		
FutureTech.Test (15)	1,3 sec		
BlobStorageServiceTests (6)	683 ms		
GenerateSasUrl_NullOrEmptyUr...	< 1 ms		
UploadImage_FileTooLarge_Fail...	1 ms		
UploadImage_FileWithinSizeLi...	1 ms		
UploadImage_InvalidFileType_T...	< 1 ms		
UploadImage_ValidJpeg_Passes...	679 ms		
Passed Image_ValidPng_PassesT...	2 ms		
StudentModelTests (9)	608 ms		

Value Proposition

Why our Secure Student Management System stands out



Secure

OAuth 2.0 + RBAC + SAS tokens provide multi-layered security for all data



Scalable

Azure auto-scaling handles growing student data without performance degradation



Easy to Use

Intuitive dashboard with clean UI makes student management effortless



Cloud-Powered

Full Azure ecosystem integration ensures reliability and global accessibility

Our system **improves efficiency, enhances security, and is ready for real-world use.**

From manual record-keeping to a **modern cloud solution.**



Thank You

Questions?

[Live Project](#)
[GitHub Repository](#)

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